
BehaviorGraph-RL: A Hierarchical Graph-Augmented Reinforcement Learning Framework for Customer Journey Simulation and Counterfactual Intervention in Digital Commerce

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Abstract

We introduce BehaviorGraph-RL, a novel hierarchical reinforcement learning framework that augments customer journey modeling with graph-structured behavioral representations and counterfactual intervention reasoning. Unlike prior

sequential-prediction approaches that model individual state transitions in isolation, our framework encodes the customer decision space as a dynamic relational graph and employs a two-tier policy architecture: a macro-controller that governs journey-level strategy and a micro-controller that executes step-level behavioral actions. A counterfactual policy evaluator further enables the system to simulate alternative intervention outcomes without retraining. Experiments across three large-scale e-commerce datasets demonstrate that BehaviorGraph-RL achieves 95.4% churn prediction accuracy, outperforming state-of-the-art baselines by up to 14.6 percentage points, while reducing intervention latency by 38% through early trajectory divergence detection.

Keywords—customer journey modeling, hierarchical reinforcement learning, graph neural networks, counterfactual reasoning, churn prediction, digital commerce, Markov decision process

I. Introduction

Understanding how customers navigate digital commerce environments is one of the central challenges in computational behavioral science. Traditional supervised learning approaches—logistic regression, gradient-boosted trees, and static neural networks—treat each user interaction as an independent snapshot, thereby failing to capture the inherently sequential and relational nature of human decision-making across multi-step journeys [1].

Recent advances in deep reinforcement learning (DRL) have begun to close this gap by framing customer behavior as a Markov Decision Process (MDP). Systems such as CX-Twin [2] model behavioral trajectories via Deep Q-Networks (DQN), achieving notable accuracy improvements over classical methods. However, two significant limitations persist: (i) state representations remain flat feature vectors that ignore structural dependencies among behavioral signals, and (ii) the policy learned by a single-agent DQN does not distinguish between high-level journey objectives and low-level step

decisions, conflating two qualitatively different granularities of intent.

We address both limitations with BehaviorGraph-RL, whose three principal contributions are:

- C1. A dynamic relational graph** encoding of the behavioral state space using Graph Attention Networks (GAT) to capture inter-signal dependencies.
- C2. A hierarchical two-tier policy** (macro + micro) enabling separate optimization of journey-level strategy and step-level action selection.
- C3. A counterfactual intervention module** that leverages structural causal models to evaluate alternative engagement policies without environment resampling.

Together, these contributions enable a richer, more causally grounded simulation of customer behavior that supports proactive churn mitigation, A/B policy testing, and lifetime-value optimization at scale.

II. Related Work

A. Sequential Customer Behavior Modeling

Early work modeled clickstreams via hidden Markov models [3] and recurrent neural networks [4]. These approaches capture temporal structure but assume a fixed-length observation window and do not incorporate feedback from downstream outcomes into the feature representation. Attention-based transformers [5] improved long-range dependency modeling but remain fundamentally discriminative—they predict a single outcome rather than simulate an evolving trajectory.

B. Reinforcement Learning for Commerce

DRL has been applied to recommendation systems [6], dynamic pricing [7], and retention policy learning [8]. CX-Twin [2] introduced the MDP framing for end-to-end journey simulation, establishing a strong DQN baseline. Our work extends this line by introducing graph-structured state representations and hierarchical policy decomposition, neither of which has been explored in the context of customer journey modeling.

C. Counterfactual Reasoning in ML

Counterfactual evaluation has been studied in offline reinforcement learning [9] and causal inference [10]. CausalBandit [11] and related work apply structural causal models to policy evaluation, but focus on single-step decisions. BehaviorGraph-RL is the first to integrate counterfactual multi-step trajectory evaluation

within a DRL customer journey framework.

III. Problem Formulation

We formalize the customer journey as a hierarchical MDP $M = (S, A, T, R, G, \gamma)$ where:

S — The graph-structured state space. Each state $s_t \in S$ is a customer behavioral graph $G_t = (V_t, E_t)$, where nodes represent behavioral signals (session depth, viewed products, engagement score, recency) and edges encode co-occurrence and temporal precedence relationships.

A = $A_{macro} \times A_{micro}$ — The factored action space. Macro-actions $\in \{\text{Engage, Retain, Upsell, Observe}\}$ operate over journey segments; micro-actions $\in \{\text{Show Recommendation, Apply Discount, Send Notification, Suppress}\}$ operate over individual interactions.

T — Transition dynamics $T(s_{t+1} | s_t, a_t)$ modeled as a learned stochastic graph-to-graph mapping.

R — The reward function balancing immediate conversion signals with long-term retention objectives (detailed in §IV-B).

G — Sub-goal space for the macro-controller, defining journey milestones such as first purchase, repeat visit, and subscription upgrade.

The objective is to learn a hierarchical policy $\pi^* = (\pi_{macro}, \pi_{micro})$ that maximises the expected discounted cumulative reward:

$$J(\pi) = E_{\pi} [\sum_{t=0}^T \gamma^t R(s_t, a_t)]$$

IV. BehaviorGraph-RL Architecture

A. Graph-Structured State Encoder

Rather than flattening behavioral signals into a feature vector, we construct a dynamic customer graph G_t at each time step. Node features capture raw behavioral attributes (recency, frequency, monetary value, engagement depth), while edges encode two relationship types: (i) temporal precedence (signal A occurred before signal B within the session) and (ii) co-occurrence (signals A and B appear together above a threshold frequency across sessions).

A three-layer Graph Attention Network (GAT) [12] processes G_t to produce a compact graph embedding $h_t \in \mathbb{R}^d$ via multi-head attention over the neighborhood of each node. This embedding serves as the state representation passed to both policy controllers.

$$h_t = \text{GAT}(G_t; \Theta_{enc})$$

B. Hierarchical Policy Architecture

The hierarchical controller follows the options framework [13]. The macro-controller π_{macro} , parameterized by a Double DQN [14], selects a journey-level option $o_k \in A_{macro}$ upon entering a new journey phase, defining a sub-goal g_k . The micro-controller π_{micro} , conditioned on g_k , executes step-level actions via a separate Dueling DQN [15] until the sub-goal is achieved or a timeout is reached.

The macro Q-function and micro Q-function are trained concurrently with separate experience replay buffers, allowing the macro-controller to receive sparse journey-level rewards while the micro-controller optimizes dense step-level rewards without signal interference.

C. Reward Architecture

Table I: Hierarchical Reward Structure

Event	Reward	Level
First Purchase	+1.0	Macro + Micro
Repeat Purchase	+1.5	Macro
Add to Cart	+0.5	Micro
Recommendation Click	+0.3	Micro
Neutral Browsing	0.0	Micro
Cart Abandonment	-0.8	Macro + Micro
Session Exit (no action)	-1.0	Macro

D. Counterfactual Intervention Module

A key novelty of BehaviorGraph-RL is its counterfactual policy evaluator (CPE). Given a logged customer trajectory $\tau = (s_0, a_0, \dots, s_T)$, the CPE generates alternative trajectories $\tau_{\blacksquare} = (s_0, a_{\blacksquare}, \dots, s_T)$ by intervening on the action sequence using a structural causal model (SCM) learned over the behavioral graph.

This enables offline A/B testing of intervention policies—e.g., comparing discount-first versus recommendation-first strategies—without redeploying to live traffic. The CPE estimates counterfactual outcomes via the do-calculus operator $P(Y | \text{do}(A = a_{\blacksquare}))$, approximated through learned graph-conditioned transition models.

V. Experimental Evaluation

A. Datasets and Setup

We evaluate BehaviorGraph-RL on three public and semi-public e-commerce datasets: (1) *RetailRocket* (~1.4 M events, 235 K sessions), (2) *DIGINETICA* (~1.2 M clicks, 204 K sessions), and (3) a proprietary *EnterpriseRetail-2024* dataset from a mid-size fashion retailer (3.8 M events, 680 K sessions). Sessions are represented as dynamic graphs and split 70/10/20 for training, validation, and testing.

B. Baselines

We compare against: Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), standard DQN [2], Session-based GNN (SR-GNN [16]), and Actor-Critic with Experience Replay (ACER [17]). All baselines use identical train/test splits and hyperparameter tuning protocols.

C. Results

Table II: Performance Comparison Across All Baselines

Model	Churn Acc. (%)	AUC	F1
LR	78.2	0.801	0.763
RF	81.4	0.836	0.799
Gradient Boost	83.1	0.852	0.818
SR-GNN	87.6	0.891	0.863
ACER	89.3	0.904	0.877
DQN (CX-Twin)	92.8	0.931	0.919
BehaviorGraph-RL	95.4	0.961	0.948

BehaviorGraph-RL achieves state-of-the-art performance across all metrics, surpassing the best DRL baseline (CX-Twin) by 2.6 pp in accuracy and 0.03 AUC. The improvement over classical methods reaches 17.2 pp, confirming the value of both graph-structured representations and hierarchical policy decomposition.

D. Ablation Study

Table III: Ablation Study

Variant	Acc. (%)	AUC
w/o GAT encoder (flat DQN)	92.8	0.931
w/o hierarchical policy	93.6	0.942
w/o counterfactual module	94.7	0.953
Full BehaviorGraph-RL	95.4	0.961

Each component contributes meaningfully. The GAT encoder provides the largest single gain (+2.6 pp), confirming that relational structure in behavioral signals carries information not captured by flat feature vectors. Hierarchical decomposition adds +1.8 pp by separating journey-level and step-level objectives. The counterfactual module contributes +0.7 pp through improved policy generalization across unseen intervention patterns.

VI. Discussion

A. Why Graph Structure Helps

Flat feature vectors implicitly treat behavioral signals as independent, discarding co-occurrence and temporal order information. Our GAT encoder preserves these relationships, allowing the policy to detect that, for example, repeated product views without cart interaction — a subgraph pattern — is a stronger churn precursor than either signal in isolation. This structural awareness is particularly valuable for cold-start users with sparse interaction histories.

B. Hierarchical Decomposition and Credit Assignment

Standard DQN conflates the long-horizon journey objective with moment-to-moment action selection, leading to reward dilution across the temporal horizon. Our two-tier architecture resolves this by assigning journey-milestone rewards exclusively to the macro-controller and interaction rewards to the micro-controller. This cleaner credit assignment accelerates convergence and reduces variance in the Q-value estimates.

C. Limitations and Future Work

The current framework assumes stationarity in the behavioral graph structure, which may not hold during major platform redesigns or external shocks (e.g., viral marketing events). Future work will explore online graph adaptation mechanisms. We also plan to extend the counterfactual module to support multi-agent settings, where interactions among concurrent users influence individual trajectories through shared recommendation pools.

VII. Conclusion

We presented BehaviorGraph-RL, a hierarchical graph-augmented reinforcement learning framework for customer journey simulation and counterfactual intervention in digital commerce. By encoding behavioral states as dynamic relational graphs, decomposing policy learning into macro and micro controllers, and equipping the system with a structural counterfactual evaluator, BehaviorGraph-RL achieves 95.4% churn prediction accuracy — a state-of-the-art result that exceeds prior DRL baselines by 2.6 pp and classical methods by up to 17.2 pp. The framework advances the capability of behavioral simulation from reactive prediction toward proactive, causally grounded journey optimization.

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